

Rahul Tamanam

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SKILLS

Data Engineering: Python, R, SQL, ETL/ELT, Snowflake, Data Pipelines, Data Modeling, Cost Optimization, PySpark, Hadoop(HDFS), Hive, Impala

Databases and Tools: MySQL, SQLite3, PostgreSQL, MongoDB, Git, Docker, FastAPI

Cloud Platforms: AWS (S3, Redshift, Glue, Lambda, Athena), GCP (BigQuery, Dataflow, Cloud)

Data Analysis and Visualization: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau, Excel

AI/ML & LLM Tools : Groq, LangChain, Pinecone, Neo4j, Ollama, Anthropic Claude API, Hugging Face (PEFT, LoRA fine-tuning), vLLM, Axolotl, Weights & Biases, Scikit-learn, XGBoost, spaCy

EXPERIENCE

Student Assistant, University Housing Operations

April 2025 – Present

The University of Texas at Dallas

Richardson, TX

- Managed 100+ weekly resident requests by logging maintenance tickets, resolving housing inquiries, and coordinating check-in/check-out operations, ensuring accurate records and timely issue resolution within the housing systems
- Developed Excel-based operational data systems leveraging formula-driven automation, and dynamic data organization to track room assignments and maintenance schedules for 800+ residents, improving reporting efficiency by 25%

AI/ML Intern

January 2024 – May 2024

HMI Engineering Services

Visakhapatnam, AP

- Built a predictive maintenance prototype by cleaning and aggregating time-series sensor data, engineering failure related features, and training classification models using Python, pandas, scikit-learn, and SQL
- Analyzed model predictions to identify high risk equipment and worked with engineers to adjust maintenance thresholds, increasing average equipment uptime by 10%
- Developed failure trend visualizations and sensor drift analyses using Matplotlib and Seaborn to communicate degradation patterns across 6 months of equipment data
- Contributed to an estimated 8% reduction in annual maintenance spend by enabling earlier interventions and fewer emergency repairs

PROJECTS

CometBot | Python, FastAPI, Neo4j, Pinecone, Typescript

January 2026 - May 2026

- Built CometBot, an AI advising platform for UTD JSOM graduate students, enabling degree planning, career guidance, and skills gap analysis using React, FastAPI, Neo4j, and Pinecone with Claude-powered LLM reasoning
- Designed a hybrid deterministic + RAG architecture where a Neo4j graph enforces academic constraints and degree progression, while Pinecone retrieval grounds LLM outputs, improving reliability of course recommendations
- Replaced client-side PDF parsing with a server-side onboarding pipeline, automatically extracting structured academic data and mirroring production-grade advising workflows

Velox | React, Node/Express, SQLite, Tailwind CSS, Recharts

May 2026

- Built a macro-aware portfolio engine using FRED (GDP, Fed Funds Rate) to classify market regimes and rebalance an ETF universe monthly, benchmarked against SPY with regime-shaded performance charts
- Ran 1,000-path Monte Carlo simulations via Geometric Brownian Motion, computing p10–p90 fan charts, max drawdown, CAGR, and goal-attainment probability from 5 years of per-ticker price history
- Developed a full-stack investment platform (React, Express, SQLite) with a composite Health Score, live Yahoo Finance integration, LLM-powered assistant, and a portfolio shock simulator

Churn Intelligence System | Python, SQL, XGBoost, FastAPI, React, LLMs

February 2026 – March 2026

- Built an end-to-end churn pipeline on 30M+ behavioral events, engineering 40+ features in DuckDB (sessionization, 30/60/90-day decay, cohort segmentation) and training an XGBoost model on 860k users with 5-fold CV achieving production-grade accuracy
- Implemented a T-learner uplift model to isolate persuadable users via incremental treatment effects, layered PELT change-point detection for behavioral regime shifts, and deployed SHAP + LLM-generated retention recommendations

EDUCATION

The University of Texas at Dallas

May 2026

Master of Science in Business Analytics and Artificial Intelligence

Gayatri Vidya Parishad College of Engineering

May 2024

Bachelor of Technology in Information Technology